Impact of Iran War-Related Events

on Oil, Gold, and EUR/USD Markets

Introductory Data Science Project · Pandas · Groupby · MA/EMA · Visualization

Presentation focus

Real data, simple analysis, clear interpretation

Yas · Data Science Course

Motivation & Research Question

Why this topic matters

Motivation

Geopolitical risk can influence financial markets, especially commodities and currencies.

Research question

Are Iran war-related events associated with visible changes in oil, gold, and EUR/USD?

Scope

This is an introductory data analysis project, not a professional trading model.

Main idea

- Use real market data and a war-event dataset
- Compare normal days vs. war-event days using groupby
- Use MA and EMA indicators to describe market trends
- Visualize results and discuss limitations

Important sentence

This project analyzes association, not causation.

Project Data & Folder Structure

Clean CSV files, notebook, and generated figures

Market data

oil_clean.csv
gold_clean.csv
eurusd_clean.csv

Event data

war_master_dataset.csv
Iran war-related event dates

Generated outputs

merged CSVs
groupby summaries
PNG charts

Project folder structure

1. data/processed - CSV input and processed outputs
2. notebooks - Jupyter Notebook
3. outputs/figures - Presentation charts
4. README.md - Project description
5. requirements.txt - Python packages

Workflow

From CSV files to charts and conclusions

1. Load CSV files

pandas read_csv

2. Clean dates

convert to datetime

3. Merge data

join by date

4. Feature engineering

daily returns + MA/EMA

5. Analyze

groupby + event windows

6. Visualize

save PNG charts

Data Cleaning & Merge

Preparing the dataset for analysis

- Standardized date columns across all files
- Sorted data chronologically
- Checked missing values and duplicate rows
- Merged oil, gold, EUR/USD, and event data by date
- Calculated daily percentage returns

Key code idea

1. Convert dates to datetime
2. Merge all datasets by date
3. Add war-event marker
4. Calculate daily percentage return

Why this matters

If the dates are not clean and aligned, the later analysis and charts can be misleading.

Groupby Analysis

Comparing normal days and war-event days

Groupby helps summarize market behavior by category.

Group data by war_event

war_event = 0: normal days
war_event = 1: event days

Then calculate average returns for each market.

Market	Normal Days	War Event Days
EUR/USD	0.038%	-0.124%
Gold	0.191%	-0.495%
Oil	0.209%	-0.131%

Interpretation

The groupby result gives a quick comparison between normal market days and war-event days.

Technical Indicators: MA and EMA

Adding market trend features

Moving Average (MA)

Simple average of past prices. It is smoother and reacts slower.

Exponential MA (EMA)

Weighted average that reacts faster to recent price changes.

Parameters

Different windows were tested: 10, 20, and 50 days.

For each market:

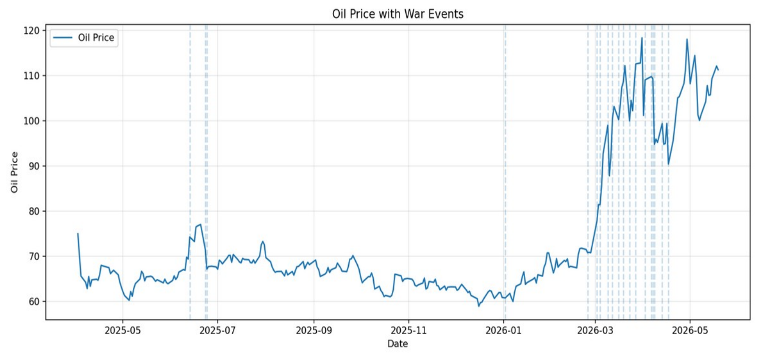
- MA 10 / 20 / 50
- EMA 10 / 20 / 50
- Bullish or Bearish trend signal

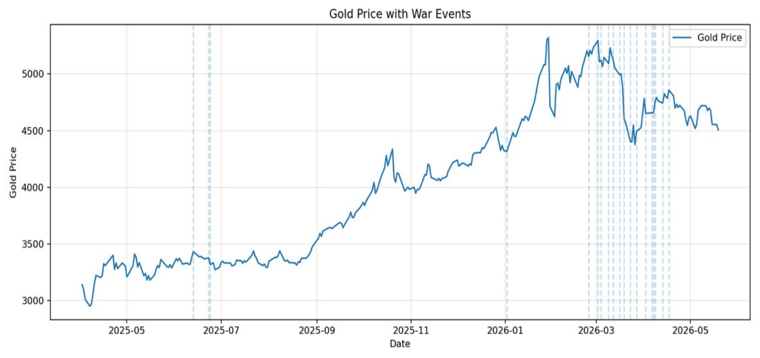
Signal idea

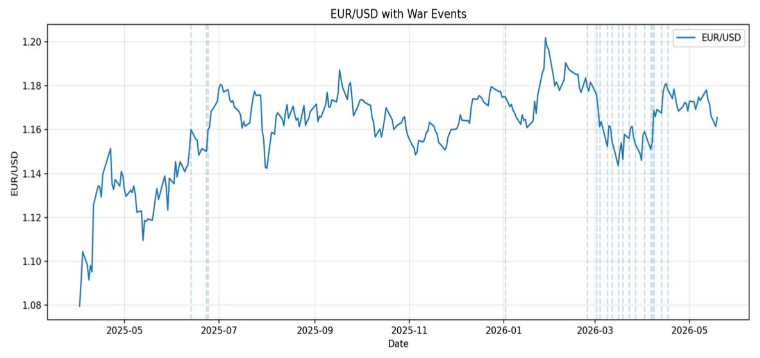
If EMA10 > EMA50 → Bullish
If EMA10 <= EMA50 → Bearish

War Event Charts

Market prices with event dates marked







Oil

Gold

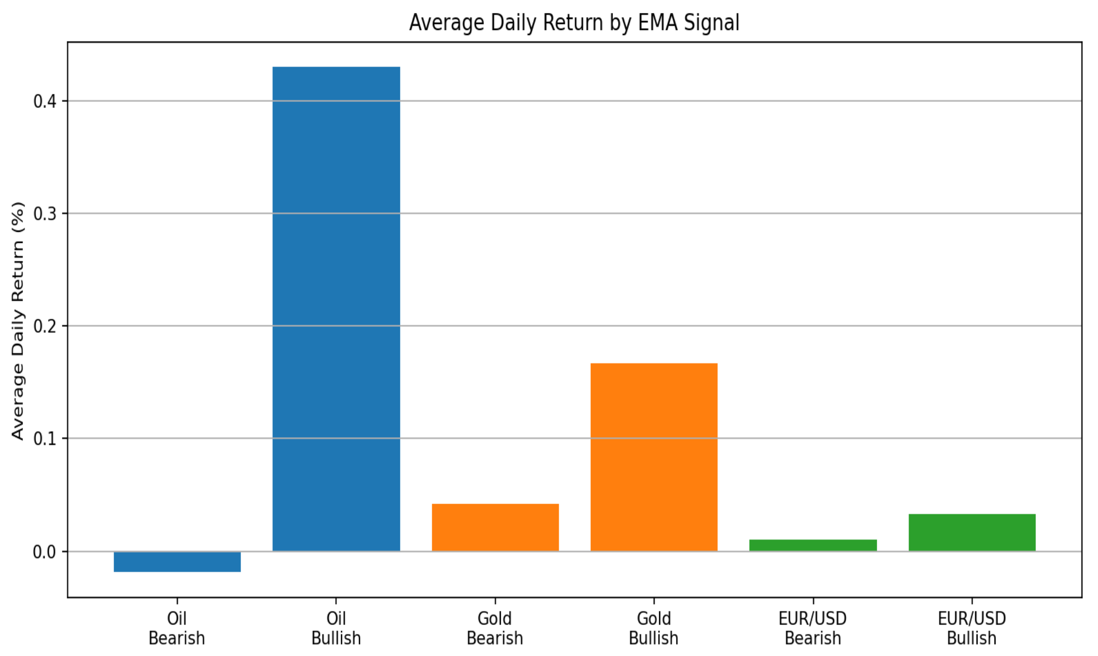
EUR/USD

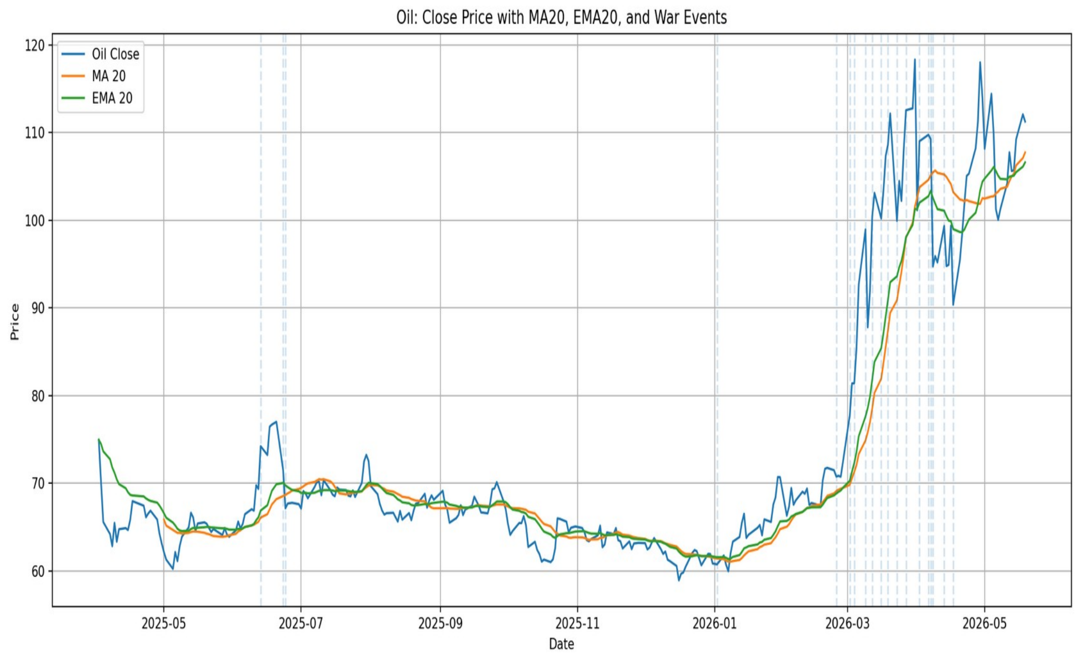
What to look at

The vertical lines mark war-related events. The charts help visually inspect market movement around these dates.

Indicator Results

EMA bullish vs. bearish conditions





EMA signal summary

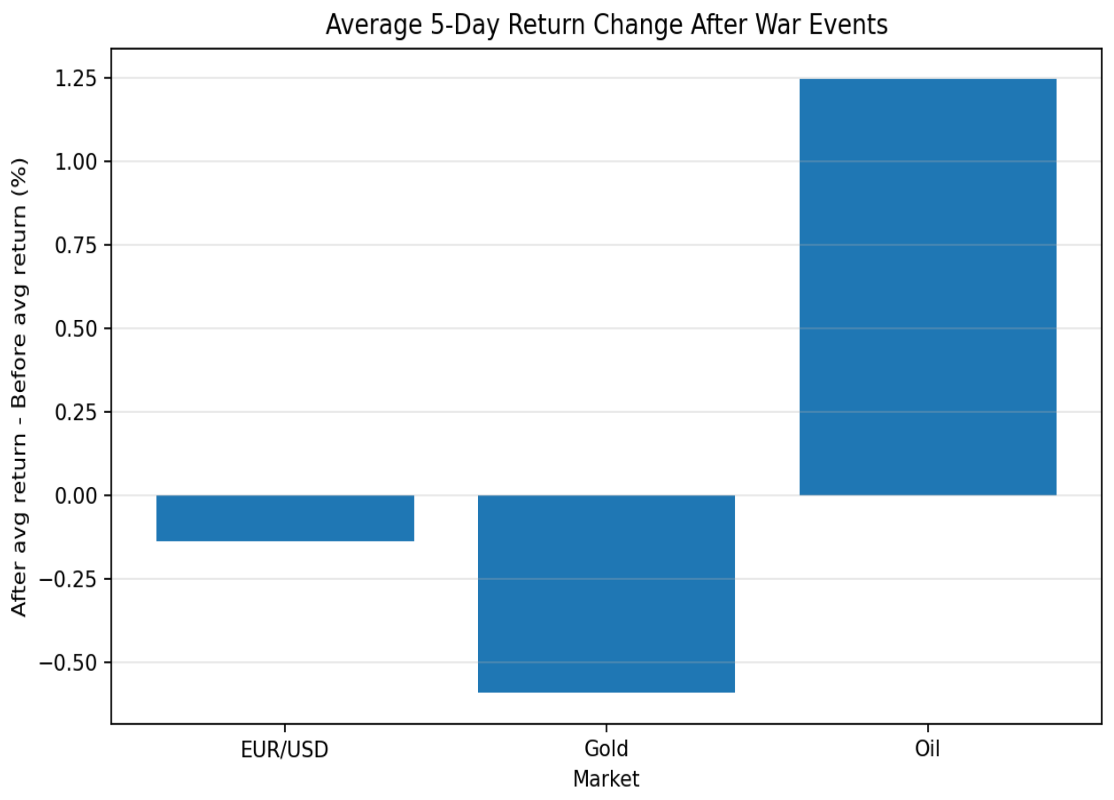
Market	Bullish	Bearish
EUR/USD	0.033%	0.011%
Gold	0.167%	0.042%
Oil	0.430%	-0.019%

Interpretation

In this dataset, EMA bullish conditions are more positive, especially for oil.

Event Window Analysis

Average 5-day change after events



Method

For each event: compare average returns 5 days before and 5 days after.

Main observation

Oil shows the strongest positive average change after events.

Careful interpretation

This indicates association in this dataset, not a proven causal effect.

Key Findings

What the analysis suggests

Oil

Strongest positive reaction in the 5-day post-event window.

Gold

Often considered a safe-haven asset, but results depend on this specific dataset.

EUR/USD

Reaction is more complex because currencies depend on many global factors.

Data Science skills demonstrated

- CSV loading and cleaning
- Merging datasets by date
- Daily return calculation
- Groupby analysis
- MA/EMA indicators
- Visual storytelling

Main conclusion

Markets show visible changes around geopolitical events, but the project does not prove causation.

Limitations & Conclusion

Simple, complete, and explainable

The dataset contains a limited number of war events

- Markets are influenced by many other factors
- Association does not prove causation
- More advanced statistical tests could improve the analysis

Final message

The project is not a prediction model. It is a clear data analysis project that connects real market data, events, indicators, and visual conclusions.

Thank you

Questions?